Jung Minyong

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Research Interset

Cosmological simulations; dwarf galaxies; ultra-faint dwarfs; machine learning; disk formation; direct N-body methods; star cluster formation

Education

Seoul National University, Seoul Mar. 2017 – Feb. 2021
Bachelor of Science - Double Major in Physics Education and Astronomy

Seoul National University, Seoul Mar. 2021 – Present
MS/PhD Student in Physics

Academic Positions

Flatiron Institute, New York, NY Jan. 2026 – May. 2026
Guest Researcher / Pre-Doctoral Researcher, Center for Computational Astrophysics (CCA)

Primary Refereed Publications

FOGGIE XVI: Ram-pressure stripping and absorption signatures of dwarf-satellite gas in Milky Way-like halos. *submitted to ApJ*
M. Jung[*], A. Wright, S. Tonnesen, J. Zhu, and other 6 co-authors

The *AGORA* High-resolution Galaxy Simulations Comparison Project. VIII: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$. *ApJ. 994, 245 (2025)*
M. Jung[*], J. -H Kim[*], T. H. Nguyen[*], R. Rodríguez-Cardoso[*], and other 21 co-authors for the *AGORA* Collaboration

Merger-tree-based Galaxy Matching: A Comparative Study across Different Resolutions. *ApJ. 965, 156 (2024)*
M. Jung[*], Kim, J. -H[*], B. K. Oh, S. E. Hong, J. Lee, and J. Kim

The *AGORA* High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo. *ApJ. 964, 123 (2024)*
M. Jung[*], S. Roca-Fàbrega[*], J. -H Kim[*], and other 18 co-authors for the *AGORA* Collaboration

The *AGORA* High-resolution Galaxy Simulations Comparison Project. IX-Part 1: Effects of a Major Galaxy Merger on Star Formation of a Milky Way-mass Galaxy Progenitor. *arXiv:2607.21709*
T. H. Nguyen[*], K. S. S. Barrow[*], **M. Jung**[*], R. Rodríguez-Cardoso[*], and 22 other co-authors for the *AGORA* Collaboration

The *AGORA* High-resolution Galaxy Simulations Comparison Project. IX-Part 2: Effects of a Major Galaxy Merger on the Stellar Morphology of a Milky Way-mass Galaxy Progenitor. *arXiv:2607.21710*
T. H. Nguyen[*], K. S. S. Barrow[*], **M. Jung**[*], R. Rodríguez-Cardoso[*], and 22 other co-authors for the *AGORA* Collaboration

The AGORA High-resolution Galaxy Simulations Comparison Project. X: Formation and Evolution of Galaxies at the High-redshift Frontier. *Accepted to ApJ, arXiv:2511.04435*
H. Kim[*], J. -H Kim[*], **M. Jung**[*], J. -H Kim[*], S. Roca-Fàbrega[*], and other 20 co-authors for the *AGORA* Collaboration

Evolution of Nuclear Star Cluster in Dwarf Galaxy through Mergers and In-Situ Star Formation. *Accepted to ApJ, arXiv:2603.18242*
Y. Jo[*], **M. Jung**, S. Kim, G. L. Bryan, J. -H Kim, and Lee, A.

The *AGORA* High-resolution Galaxy Simulations Comparison Project. VII: Satellite quenching in zoom-in simulation of a Milky Way-mass halo. *A&A. 698. A303*
R. Rodríguez-Cardoso[*], S. Roca-Fàbrega[*], **M. Jung**[*], T. H. Nguyen[*], and other 18 co-authors for the *AGORA* Collaboration

The *AGORA* High-resolution Galaxy Simulations Comparison Project. IV: Halo and Galaxy Mass Assembly in a Cosmological Zoom-in Simulation at $z \leq 2$. *ApJ. 968, 125 (2024)*
S. Roca-Fàbrega[*], J. -H Kim[*], J. R. Primack[*], **M. Jung**[*], and other 23 co-authors for the *AGORA* Collaboration

Dark Matter Deficient Galaxies Produced Via High-velocity Galaxy Collisions In High-resolution Numerical Simulations. *ApJ. 899, 25 (2020).*
E. -J. Shin[†], **M. Jung**[†], G. Kwon[†] , J. -H Kim[*], J. Lee, Y. Jo, B. K. Oh

[*]corresponding author
[†]these authors contributed equally to this work

Other Refereed Publications

From Dense Gas Clouds to Supermassive Black Hole Seeds: Hybrid Hydro/Direct N-body Simulations of Runaway Collision-driven Intermediate-mass Black Hole Formation. *ApJ. 1004, 250 (2025)*
E. Chung, Y. Jo, J. -H Kim, **M. Jung**, and O. Kwon

Inferring Cosmological Parameters on SDSS via Domain-generalized Neural Networks and Light-cone Simulations. *ApJ. 975, 38 (2024)*
J. -Y Lee, J. -H Kim*, **M. Jung**, and other 6 co-authors

The *AGORA* High-resolution Galaxy Simulations Comparison Project. VI: Similarities and Differences in the Circumgalactic Medium. *ApJ. 962, 29 (2024)*
S. Strawn*, S. Roca-Fàbrega*, J. R. Primack*, J. -H Kim*, and other 28 co-authors including **M. Jung**

Awards

Outstanding Presentation Award — Apr. 23, 2023
Oral Session (#) — *The Korean Physical Society*

Outstanding Project Award (1st Place) — June. 30, 2023
The 14th KIAS CAC Summer School on Artificial Intelligence & Parallel Computing (#) — *Korea Institute for Advanced Study*

Outstanding Project Award (3rd Place) — Feb. 24, 2024
2024 Winter School on Programming for Accelerators - Advanced Course (#) — *SNU THUNDER Research Group*

Oral Presentaions

Numerical Galaxy Formation Mini-Workshop (#) — Jan. 16. 2020
Dark matter deficient galaxies produced via high-velocity galaxy collisions in high-resolution numerical simulations — *Shin, E. -J & M. Jung*

2023 Darwin+NGF Workshop (#) — Jan. 11. 2023
The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo — *M. Jung*

2023 KPS spring meeting (#) — Apr. 21. 2023
The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo — *M. Jung*

XV International Conference on Gravitation, Astrophysics and Cosmology (ICGAC15, #) — Jul. 3. 2023
The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo — *M. Jung*

2023 Santa Cruz Galaxy Workshop (#) — Aug. 10. 2023
The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo — *M. Jung*

SNU ARC 2nd H.S. Yun Astronomy Workshop (#) — Aug. 30. 2023
The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo — *M. Jung*

2023 108th KAS Fall Meeting (#) — Oct. 18. 2023
Merger-tree-based Galaxy Matching: A Comparative Study across Different Resolutions — *M. Jung*

The 2nd Workshop on Galaxies and Dark Matter (Hosted by SNU LAMP Foundation) — Feb. 27. 2024
The AGORA High-resolution Galaxy Simulations Comparison Project: A MW-mass Galaxy and Substructures in a Cosmological Zoom-in Simulation — *M. Jung*

The 2nd CTP Bosan Workshop: AGORA in Asia + 5th Numerical Galaxy Formation Meeting in Korea (#) — May. 7. 2024
The Satellite Galaxy Population in the AGORA CosmoRun, and Resolution Convergence Test in the TNG Simulation — *M. Jung*

2024 110th KAS Fall Meeting (#) — Oct. 16. 2024
The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$ — *M. Jung*

The 3rd Workshop on Galaxies and Dark Matter (#) — Feb. 27. 2025
The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$ — *M. Jung*

2025 Santa Cruz Galaxy Workshop (#) — Aug. 8. 2025
The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$ — *M. Jung*

SNU ARC 4nd H.S. Yun Astronomy Workshop (#) — Aug. 21. 2025
The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$ — *M. Jung*

CCA/NY Area Galaxy Formation Seminar — Jan. 30. 2026
Star Clusters and Galaxies Using Self-consistent Hybrid Hydro/direct N-body Simulations — *M. Jung*

2026 CCA Pre-Doc Symposium (#) — May. 19. 2026
Fate of ram-pressure stripped gas from resolved dwarf galaxies — *M. Jung*

1st Galaxy Workshop and AGORA in Europe (#) — May. 27. 2026
Fate of ram-pressure stripped gas from resolved dwarf galaxies — *M. Jung*

Poster Presentations

2023 DARWIN-Dwarf galaxy researcher workshop (#) Aug. 16, 2023
Merger-tree-based Galaxy Matching: A Comparative Study across Different Resolutions *M. Jung*

The 11th KIAS Workshop on Cosmology & Structure Formation (#) Oct. 27, 2024
The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$ *M. Jung*

Experiences

Peer Reviewer 2025 – Present
American Astronomical Society Journals

- Reviewed manuscripts as an invited referee

Undergraduate Internship Mar. 2019 – Aug. 2020
Department of Physics & Astronomy (Advisor: Prof. Kim Ji-hoon) *SNU*

- Studied cosmological hydrodynamical simulations
- Investigated the kinematics of dark matter and baryons in galaxies via simulations
- Conducted a statistical analysis of the spin of galaxies

System Administrator Sep. 2021 – Present
Computational Cosmology Group (Advisor: Prof. Kim Ji-hoon) *SNU*

- Administered the lab's computing cluster with root access; managed user environments, installed scientific software, and ensured system stability

Teaching Experiences

Assistant Mentor, 1 year, SNU Science Education Institute for The Gifted, Student Research Mentoring Program (Physics) 2022
Teaching Assistant, 1 semester, Seoul National University, Rudimentary Mathematical Methods of Physics 2022
Teaching Assistant, 1 semester, Seoul National University, Physics 1 2022
Teaching Assistant, 1 semester, Seoul National University, Rudimentary Mathematical Methods of Physics 2021
Teaching Assistant, 1 semester, Seoul National University, Physics Lab.1 2021
Teaching Practicum, 1 semester, Seoul National University Middle School 2020

Collaborations and Technical Skills

Collaborations: AGORA, FOGGIE
Programming Languages: Python, C, C++, CUDA
Programs: ENZO, GADGET-4, MUSIC
Libraries: yt, scipy, pandas, pytorch
Open-Source Contributions: ABYSS, ENZO-ABYSS